SoSe 26 Type Theory & Logic & Functional Programming Notes

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Preface

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Part I

Logic

1 Lecture 1 - Propositional Logic: Syntax

1.1 Overview

Brief summary of what this lecture does (2–4 lines).

Goal: Define the expression we can write, these are called **well-formed-formulas**, or **wff**s.

There are various ways of defining what counts as a wff. One of them is the inductive approach.

1.2 Inductive Approach

First we start with **atomic formulas**:

$$P_0, P_1, P_2, ...$$
$$(P_i)_{i \in \mathbb{N}}$$

are atomic propositions.

We give some rules to inductively define new formulas from old ones.

1) if φ, ψ are **wff** then $\varphi \Rightarrow \psi$ is a **wff**
2) if φ, ψ are **wff** then $\varphi \wedge \psi$ is a **wff**
3) if φ, ψ are **wff** then $\varphi \vee \psi$ is a **wff**
4) if φ is a **wff** then $\neg\varphi$ is a **wff**

1.3 Constructor Approach

It is useful to express these rules with **constructors**, which define wffs as values of some functions (with codomain WFF)

In this approach atomic formulas are values of functions:

Definition 1.1 (Atomic formulas)**.** Atomic formulas are values of functions like:

$$P : \mathbb{N} \to \text{WFF}, \quad : i \mapsto P_i$$

or

$$''\cdot'' : \text{String} \to \text{WFF}, \quad : x \mapsto "x"$$

Example 1.1 (String Constructor)**.** with the latter we can construct arbitrary propositions like:

"today␣is␣sunny"

We also have constructors for the logical connectives:

Definition 1.2 (Constructors for Logical Connectives)**.**

$$\begin{aligned}
(\neg) &: \text{WFF} \to \text{WFF}, \quad : \varphi \mapsto \neg\varphi \\
(\wedge) &: \text{WFF} \times \text{WFF} \to \text{WFF}, \quad : (\varphi, \psi) \mapsto \varphi \wedge \psi \\
(\vee) &: \text{WFF} \times \text{WFF} \to \text{WFF}, \quad : (\varphi, \psi) \mapsto \varphi \vee \psi \\
(\Rightarrow) &: \text{WFF} \times \text{WFF} \to \text{WFF}, \quad : (\varphi, \psi) \mapsto \varphi \Rightarrow \psi
\end{aligned}$$

Remark 1.1 (Constructor vs Symbol View of Logical Connectives). Earlier we wrote $\varphi \Rightarrow \psi$ and then we denoted $\Rightarrow$ symbol as a function of type $\text{WFF} \times \text{WFF} \to \text{WFF}$, we use $(\Rightarrow)$ to distinguish between the two concepts.

A different but related question that arises si the ambiguity of $P_1 \Rightarrow P_2 \Rightarrow P_3$. According to the inductive definition there are two ways this formula could have been constructed which corresponds to the different formulas:

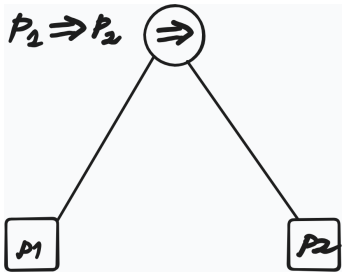
1. $(P_1 \Rightarrow P_2) \Rightarrow P_3$
2. $P_1 \Rightarrow (P_2 \Rightarrow P_3)$

To resolve this ambiguity we define the associativity rule:

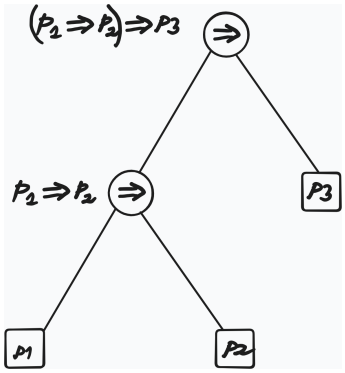
Definition 1.3 (Associativity Rule for $\Rightarrow$)**.**

$$P_1 \Rightarrow P_2 \Rightarrow P_3 := (P_1 \Rightarrow P_2) \Rightarrow P_3$$

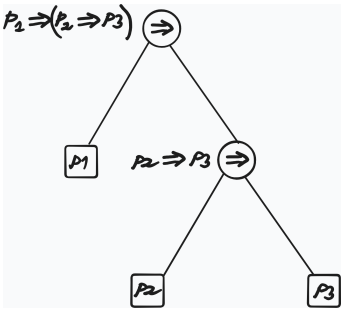
Remark 1.2 (Wffs as Trees). Wffs can be viewed as trees. Trees encode the derivation structure and resolve the ambiguity



(a) tree 1



(b) tree 2



(c) tree 3

our collection of wffs can be further enriched with another binary constructor:

$$\begin{aligned}(\Leftrightarrow) &: \mathrm{WFF} \times \mathrm{WFF} \to \mathrm{WFF} \\ &: (\varphi, \psi) \mapsto \varphi \Leftrightarrow \psi\end{aligned}$$

Remark 1.3 (Meaning of $\Leftrightarrow$). At this stage there is no relation between

$$\varphi \Leftrightarrow \psi \quad \text{and} \quad (\varphi \Rightarrow \psi) \wedge (\psi \Rightarrow \varphi)$$

Let is give another 'set-theoretic' approach

1.4 Set Theoretic Approach

First we define the alphabets:

Definition 1.4 (Set Theoeretic Approach to Defining Wffs)**.** We start with the alphabets:

$$\begin{array}{ll}\mathcal{A} := \{P_i\}_{i\in\mathbb{N}} & \text{(sub-alphabet of atomic formulas)} \\ \mathcal{C} := \{\Rightarrow, \wedge, \vee, \neg, \Leftrightarrow\} & \text{(sub-alphabet of logical connectives)} \\ \mathcal{V} := \mathcal{A} \cup \mathcal{C} \cup \{(,)\} & \text{(alphabet of propositional logic, including punctuation marks)}\end{array}$$

Then we define the so-called **Kleene Closure** of $\mathcal{V}$:

Definition 1.5 (Kleene Closure of $\mathcal{V}$)**.**

$$\mathcal{V}^* := \bigcup_{i=0}^{\infty} \mathcal{V}_i$$

$\mathcal{V}^*$ is the language containing all sorts of strings that can be constructed from the alphabet $\mathcal{V}$, also senseless ones.

To define the language propositional logic that contains only the formulas that we want and nothing else, we take the usual set-theoretic approach of taking the infinite union of all sets $\mathcal{W} \subseteq \mathcal{V}$ that satisfy the following properties:

1) $\mathcal{A} \subseteq \mathcal{W}$
2) if $\varphi, \psi \in \mathcal{W}$ then $\varphi \diamond \psi \in \mathcal{W}$, where $\diamond \in \{\Rightarrow, \wedge, \vee, \Leftrightarrow\}$
3) if $\varphi \in \mathcal{W}$ then $\neg\varphi \in \mathcal{W}$

Then the language of propositinal logic is defined as:

Definition 1.6 (Language Propositional Logic as Infinite Intersection)**.**

$$\mathcal{F}(\mathcal{A}) := \bigcap \{\mathcal{W} \subseteq \mathcal{V}^* | \mathcal{W} \text{ satisfies (1), (2), and (3)}\}$$

Theorem 1.1.

With this construction $\mathcal{F}(\mathcal{A})$ is the smallest set that satisfies the properties (1), (2) and (3), i.e.

1) $\mathcal{F}(\mathcal{A})$ satisfies the properties (1), (2) and (3)
2) For any set $\mathcal{W}$ that satisfies the properties (1), (2) and (3) we have $\mathcal{W} \subseteq \mathcal{F}(\mathcal{A})$

1.5 Summary

- WFF is defined inductively

- Constructors that generate WFFs
- Set theoretic construction of WFFs

2 Lecture 2 - Height and Subformulas

2.1 Overview

- We provide a fourth, iterative set construction for the language of propositional logic.
- We give two equivalent definitions for the height of a formula.
- We give two equivalent definitions of a function that gives the list of subformulas of a formula

2.2 Recurisve Set Construction

We define $\mathcal{F}_0$ iteratively as follows:

$$\begin{aligned}
&\mathcal{F}_0 := \mathcal{A} \\
&\mathcal{F}_1 := \mathcal{F}_0 \cup \{\, \neg\varphi \mid \varphi \in \mathcal{F}_0 \,\} \cup \{\, \varphi \diamond \psi \mid \varphi, \psi \in \mathcal{F}_0 \,\} && (\diamond \in \mathcal{C}) \\
&\mathcal{F}_n := \mathcal{F}_{n-1} \cup \{\, \neg\varphi \mid \varphi \in \mathcal{F}_{n-1} \setminus \mathcal{F}_{n-2} \,\} \cup \{\, \varphi \diamond \psi \mid \varphi, \psi \in \mathcal{F}_{n-1} \,\} && (n \geq 2)
\end{aligned}$$

With this construction it follows that:

$$\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$$

Example 2.1. $\neg\neg P_4 \in \mathcal{F}_2 \subseteq \mathcal{F}_3 \subseteq ...$

but $\neg\neg P_4 \notin \mathcal{F}_1$, because $\neg P_4 \notin \mathcal{F}_0$

Now we make the following definition:

Definition 2.1.

$$\mathcal{F} := \bigcup_{n \in \mathbb{N}} \mathcal{F}_n$$

and we claim:

Lemma 2.1.

$$\mathcal{F} = \mathcal{F}(\mathcal{A})$$

Where $\mathcal{F}(\mathcal{A})$ is defined in Definition 1.6

Proof. To prove $\mathcal{F} \subseteq \mathcal{F}(\mathcal{A})$, it suffices to show that

$$\mathcal{F}_n \subseteq \mathcal{F}(\mathcal{A}), \quad \forall n, \text{ by induction on } n$$

To conclude the proof we need to show that $\mathcal{F}_A \subseteq \mathcal{F}$ □

Definition 2.2 (Height of Formula - Set Approach)**.** For all $\varphi \in \mathcal{F}$, there is a well defined natural number $\mathrm{ht}(\varphi) \in \mathbb{N}$:

$$\mathrm{ht}(\varphi) := \min\{\, n \in \mathbb{N} \mid \varphi \in \mathcal{F}_n \,\}$$

called the **height** of φ

We give another, a more computational definition for the height.

Definition 2.3 (Height of a Formula - Computational Approach)**.** We define $\mathrm{ht}(\cdot)$ recursively on the structure of a formula ϕ as follows:

$$\begin{aligned}
&\mathrm{ht}(P_i) = 0 && (P_i \in \mathcal{A}) \\
&\mathrm{ht}(\neg\varphi) = 1 + \mathrm{ht}(\varphi) \\
&\mathrm{ht}(\varphi \diamond \psi) = 1 + \max(\mathrm{ht}(\varphi), \mathrm{ht}(\psi)) && (\diamond \in \mathcal{C})
\end{aligned}$$

Remark 2.1 (Pattern Matching). Such inductive (recursive) definitions are also called **pattern matching**

Example 2.2 (Derivation of Height)**.**

$$\begin{aligned}
\mathrm{ht}(\neg\neg(P_1 \Rightarrow P_2)) &= 1 + \mathrm{ht}(\neg(P_1 \Rightarrow P_2)) \\
&= 1 + 1 + \mathrm{ht}(P_1 \Rightarrow P_2) \\
&= 1 + 1 + \max(\mathrm{ht}(P_1), \mathrm{ht}(P_2)) \\
&= 1 + 1 + \max(1, 1) \\
&= 1 + 1 + 1 \\
&= 3
\end{aligned}$$

Obervation:

We observe that the height of a formula is the height of its syntax tree

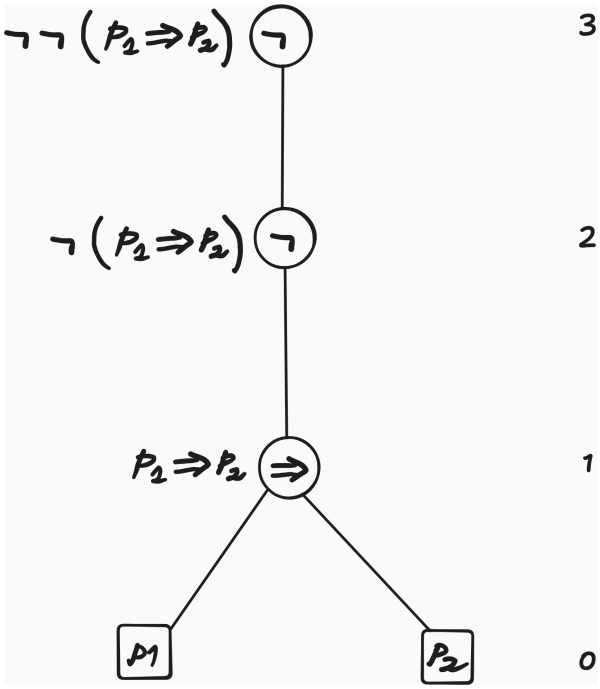


Figure 2.1: height of the syntax tree

We want to define a function

$$\mathrm{sf} : \mathcal{F}(\mathcal{A}) \to \mathcal{P}(\mathcal{F}(\mathcal{A}))$$

mapping a formula to its set of subformulas. For this we are going to define functions $\mathrm{sf}_i : \mathcal{F}_i \to \mathcal{P}(\mathcal{F})$ as follows:

Definition 2.4 (Set Construction of sf)**.**

$$\begin{aligned}
&\mathrm{sf}_0(P_i) := \{P_i\} \\
&\mathrm{sf}_n(\neg\varphi) := \{\neg\varphi\} \cup \mathrm{sf}_{n-1}(\varphi) \\
&\mathrm{sf}_n(\varphi \diamond \psi) := \{\varphi \diamond \psi\} \cup \mathrm{sf}_{n-1}(\varphi) \cup \mathrm{sf}_{n-1}(\psi)
\end{aligned}$$

Then sf can be defined as

$$\begin{aligned}
\mathrm{sf} &: \mathcal{F} \to \mathcal{P}(\mathcal{F}) \\
&: \varphi \mapsto \mathrm{sf}_{\mathrm{ht}(\varphi)}(\varphi)
\end{aligned}$$

We give another, more computational approach, by defining sf recursively on the structure of wffs as a function $\mathrm{sf} : \mathrm{WFF} \to [\mathrm{WFF}]$

Definition 2.5 (List Construction of sf)**.**

$$\begin{aligned}
&\mathrm{sf}(P_i) := [P_i] \\
&\mathrm{sf}(\neg\varphi) := [\neg\varphi] +\!\!+ \mathrm{sf}(\varphi) \\
&\mathrm{sf}(\varphi \diamond \psi) := [\varphi \diamond \psi] +\!\!+ \mathrm{sf}(\varphi) +\!\!+ \mathrm{sf}(\psi)
\end{aligned}$$

Example 2.3 (Derivation of the List of Subformulas with sf)**.**

$$\begin{aligned}
\mathrm{sf}(P_1 \vee \neg P_2) &= [P_1 \vee \neg P_2] +\!\!+ \mathrm{sf}(P_1) +\!\!+ \mathrm{sf}(\neg P_2) \\
&= [P_1 \vee \neg P_2] +\!\!+ [P_1] +\!\!+ [\neg P_2] +\!\!+ \mathrm{sf}(P_2) \\
&= [P_1 \vee \neg P_2] +\!\!+ [P_1] +\!\!+ [\neg P_2] +\!\!+ [P_2] \\
&= [P_1 \vee \neg P_2, P_1, \neg P_2, P_2]
\end{aligned}$$

2.3 Summary

- iterative set construction $\mathcal{F}_i$ - the language of propositional logic as infinite union of such
- two definitions for ht - height of a formula
- two definitions for sf - list or set of subformulas of a formula

References